# Hsin-Yu Lai

Software Engineer

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## SUMMARY

Software Engineer with 3+ years at MathWorks building features across the full stack — frontend UI systems, standalone widgets, backend tooling, and cross-platform mobile integration — for products used by 5M+ MATLAB users. Leverages LLMs (Claude) daily to accelerate development and debugging, and to generate unit and end-to-end test cases. Proven track record of driving cross-team initiatives with measurable results: ~90% memory reduction in docking workflows, ~1 month cut in cross-platform integration time, and 40% reduction in bug escape rate.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, Java, C++

**Frontend:** React, React Native, Redux, Next.js, Angular, HTML5, CSS3

**Backend:** Node.js, Express, Flask, Django

**Database & Cloud:** MongoDB, MySQL, PostgreSQL, DynamoDB, Firebase, AWS S3, GCP, Docker

**AI Tools:** LLM-assisted development (Claude), LLM-generated unit & E2E tests, prompt engineering

## PROFESSIONAL EXPERIENCE

**MathWorks** — Software Engineer Jun 2022 – Apr 2026

- Developed and shipped features for the MATLAB Live Editor and Code Editor (used by 5M+ users) using JavaScript and TypeScript, improving editor rendering speed and UI responsiveness.
- Built a feature within the docking/undocking system and led its testing — managing complex UI state across panel states, layout changes, and dynamic resize behavior — identifying and resolving performance bottlenecks to reduce memory usage by ~90% (verified via Memlab).
- Designed and built the frontend of standalone widgets (including a live code comparison widget) embedded across multiple MATLAB products — owning UI architecture, component design, and integration with backend services.
- Built and maintained backend tooling for MATLAB Mobile cross-platform integration, driving CI/CD pipeline automation that cut integration time by ~1 month and reduced bug escape rate by 40% by shifting detection earlier in the cycle.
- Led the design, delivery, and end-to-end testing of Presentation Mode for the MATLAB Editor — enabling slideshow-style presentations directly within the editor — driving alignment across engineering and UX and reaching ~80% code coverage while keeping flaky failures under 10%.

**MathWorks** — Engineer, Engineering Development Group Feb 2022 – Jun 2022

- Resolved 50+ support cases across 4+ product areas — MATLAB Editor, installation, licensing, and MATLAB Online — through systematic root-cause analysis and rapid domain knowledge acquisition.
- Identified issue impact and scope quickly, providing technical guidance to engineering teams to reduce recurring issues.

## PROJECTS

**Siitch App** — iOS Carbon Footprint Tracking App Sep 2020 – Jan 2021

- Built a full-featured iOS app with React Native and Firebase to track daily carbon footprint — covering activity logging, real-time data sync, and emission statistics charts.

## EDUCATION

**University of Southern California** — M.S. Computer Science GPA 3.75/4.0 | 2020 – 2021

**Univ. of Illinois at Urbana-Champaign** — M.S. Materials Science GPA 4.0/4.0 | 2019

**National Cheng Kung University** — B.E. Resources Engineering GPA 3.68/4.0 | 2014 – 2018